



**Shri Vile Parle Kelavani Mandal's
USHA PRAVIN GANDHI COLLEGE OF ARTS,
SCIENCE AND COMMERCE (Autonomous)**



AI-RESCUE: INTELLIGENT DISASTER RESPONSE & RESOURCE COORDINATION PLATFORM

**A Field Project Report
Submitted in Partial Fulfillment
For Assessment in
Third Year
BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)**

Submitted by

Name: Dev Vikam

SAP ID: 53013240013

**Under the esteemed guidance of
Dr. Manisha Divate**

CERTIFICATE

This is to certify that Mr. Dev Vikam, SAP ID 53013240013, has successfully completed the Field Project Work in B.Sc. Information Technology under the Department of Information Technology with the project title "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" under the supervision and guidance of Dr. Manisha Divate.

The project focuses on the design and development of an AI-powered emergency disaster response and resource coordination platform that enables multi-channel citizen incident reporting, automated SOS alerting, multi-modal severity triage, dynamic GIS responder mobilization, hospital casualty transfer coordination, and shelter evacuee capacity tracking.

We further certify that the work has been carried out by the student under proper academic guidance and that the project report has been prepared for academic assessment as per the requirements of the B.Sc. Information Technology curriculum.

Dr. Manisha Divate
Project Guide

Course Coordinator /
HOD

Principal
UPG College

Date: _____

External Examiner

DISASTER RESPONSE & CRISIS TECH LABS

Center for Disaster Informatics, AI Solutions & Emergency Coordination
Mumbai, Maharashtra - 400056

COMPANY CERTIFICATE

This is to certify that Mr. Dev Vikam, a student of Third Year B.Sc. Information Technology (SAP ID: 53013240013) from Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), has successfully undergone and completed his Field Project work on the project entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" from June 2026 to September 2026 under the mentorship and technical supervision of our technical engineering team.

During this 90-hour engagement (30 hours guide-supervised reviews and 60 hours independent development work), he demonstrated excellent analytical capabilities, technical proficiency in full-stack architecture, spatial query algorithms, and high commitment to real-time humanitarian engineering solutions. His conduct and performance throughout the tenure were exemplary.

Date: 26th September 2026

Place: Mumbai

Authorized Signatory
Technical Director / HR
Disaster Response & Crisis Tech Labs

DECLARATION

I, Mr. Dev Vikam, hereby declare that the project report entitled "AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform" represents my original work carried out under the guidance of Dr. Manisha Divate.

The work presented in this report has been completed as part of the academic requirements of the Bachelor of Science in Information Technology (Third Year, Semester V) curriculum of Usha Pravin Gandhi College of Arts, Science and Commerce (Autonomous), affiliated to University of Mumbai.

I further declare that this report has not been submitted previously to any other university, institute, or examining body for the award of any degree, diploma, or certificate.

Date: 26th September 2026
Place: Mumbai

Mr. Dev Vikam
SAP ID: 53013240013

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LOG BOOK

Field Project – 90 Hours

Project Title: AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform

GUIDE SESSIONS – 30 HOURS

Total: 30 hours guide-supervised project work

Date	Hours	Activity
12/06/2026	3 hours	Project orientation and AI-Rescue overview; understanding the problem of fragmented disaster response and defining project scope.
18/06/2026	3 hours	Requirement gathering discussion: citizen incident reporting, SOS alerts, responder mobilization, and resource allocation requirements.
20/06/2026	3 hours	Review of proposed technology stack: Next.js frontend, FastAPI backend, PostgreSQL/PostGIS spatial database, and Redis caching.
25/06/2026	3 hours	Detailed UI/UX workflow discussion: citizen reporting portal, admin command center, and first responder mobile interfaces.
27/06/2026	3 hours	Resource allocation logic review: matching nearest available resources, hospitals, and shelters to high-severity incidents.
02/07/2026	3 hours	Review of functional and non-functional requirements; performance benchmarks for sub-second SOS alert dispatching.
23/07/2026	3 hours	Project schedule and milestone planning; Gantt chart construction and Work Breakdown Structure (WBS) approval.
13/08/2026	3 hours	System architecture, ER Diagram (ERD), Data Flow Diagrams (DFDs), and end-to-end System Flowchart review.
29/08/2026	3 hours	Relational database schema finalization (16 tables) and AI multi-modal severity scoring module validation.
12/09/2026	3 hours	Interim project review, documentation sign-off, system integration verification, and viva preparation.

Name & Signature of Student

Dev Vikam

Name & Signature of Teacher

Dr. Manisha Divate

LOG BOOK

Field Project – 90 Hours

Project Title: AI-Rescue: Intelligent Disaster Response & Resource Coordination Platform

WORK – 60 HOURS

Total: 60 hours independent development & research work

Date	Hours	Activity
12/06/2026	3 hours	Field study on disaster communication delays; comprehensive requirement analysis for AI-Rescue.
13/06/2026	3 hours	Literature survey of FEMA, NDMA, Ushahidi, and Sahana disaster response frameworks.
15/06/2026	3 hours	Next.js 14 and FastAPI development environment setup; Docker containerization configuration.
17/06/2026	3 hours	UI wireframing for Citizen SOS dispatch, Incident Map view, and Resource Tracker dashboards.
19/06/2026	3 hours	PostgreSQL database initialization with PostGIS spatial extensions for geo-tagging coordinate storage.
20/06/2026	3 hours	Authentication module implementation: JWT authentication, bcrypt hashing, and Role-Based Access Control.
22/06/2026	3 hours	Citizen Incident Reporting module: GPS reverse-geocoding, multi-category selection, and photo upload pipeline.
24/06/2026	3 hours	Live incident feed integration: WebSocket pub/sub implementation via Redis for real-time map updates.
26/06/2026	3 hours	AI multi-modal NLP classification pipeline: text severity scoring and OCR text extraction from disaster images.
28/06/2026	3 hours	Severity ranking and triaging engine development using fine-tuned DistilBERT / ensemble scoring rules.
03/07/2026	3 hours	Resource inventory tracking module: real-time stock levels, vehicle tracking, and specialized equipment status.
05/07/2026	3 hours	Nearest-responder dispatch algorithm: Haversine distance calculations and road network ETA estimation.
07/07/2026	3 hours	Hospital emergency ward capacity tracking: ICU beds, trauma units, and real-time bed availability updates.
09/07/2026	3 hours	Relief shelter logistics module: evacuee registration, food/medical supply inventory, and occupancy rate alerts.
10/07/2026	3 hours	Multi-channel broadcast alert system: automated SMS gateways, push notifications, and email dispatch.
11/07/2026	3 hours	Incident lifecycle management: status transitions (Reported -> Verified -> In Progress -> Resolved -> Closed).
12/07/2026	3 hours	Complete Database ERD and MySQL Workbench physical relational schema reverse engineering.
13/07/2026	3 hours	Comprehensive Data Flow Diagrams (DFD Level 0, Level 1) and UML Use Case modeling.
14/07/2026	3 hours	End-to-end System Flowchart (41 operational nodes) and UML Sequence Diagram construction.
15/07/2026	3 hours	System integration testing, test case execution (4.6), performance validation, and final report compilation.

Name & Signature of Company / Organization

Disaster Response & Crisis Tech Labs

Name & Signature of Student

Dev Vikam

Name & Signature of Teacher

Dr. Manisha Divate